

Can AI-LLM Write Your SQL? Yes. Should You Trust It? Let's Find Out.

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 - Author: Dr. Mahmoud Parsian
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Part I — When AI Gets It Wrong

1 — The Question Every Student Asks

“Why do I need to learn SQL? I can just ask AI to write it for me.”

Fair question. AI tools like ChatGPT and Claude can generate SQL. The queries they produce often look correct, run without errors, and return a table of results. So what's the problem?

The problem is that “runs without errors” and “gives the right answer” are two completely different things.

A query can execute perfectly and still return the wrong number. The wrong revenue. The wrong customer count. The wrong average. The wrong ranking. And if you don’t know SQL, you will never notice.

In this document, we will use a realistic retail database to show you seven real examples where AI-generated SQL runs successfully but produces an incorrect answer — the kind of mistake that could cost a company money, damage a report, or lead to a bad business decision.

We will also show you four examples where AI gets it right — and explain why you still need SQL knowledge to verify and trust the output.

Note: You don’t need to understand every SQL keyword in this document yet. Focus on the results — what the AI got wrong and why it matters. You will learn all of these techniques during the course.

2 — The Company: BrightCart Retail

BrightCart is a small online retailer that sells office supplies and electronics. They have been in business since 2023 and track everything in a database: customers, products, orders, line items, and employees.

You are a new business analyst at BrightCart. Your manager gives you questions about the business, and you decide to let AI write the SQL. Let’s see how that goes.

3 — The Database Schema

BrightCart’s database has five tables. Here is what each one stores:

Table	What It Stores	Key Columns
customers	People who have accounts	name, city, state, join date
products	Items for sale	name, category, current price
orders	Purchase transactions	customer, date, status (Delivered / Cancelled / Returned / Pending)
order_items	Individual line items within an order	product, quantity, price at time of sale
employees	Company staff	name, department, salary, manager

How the Tables Connect

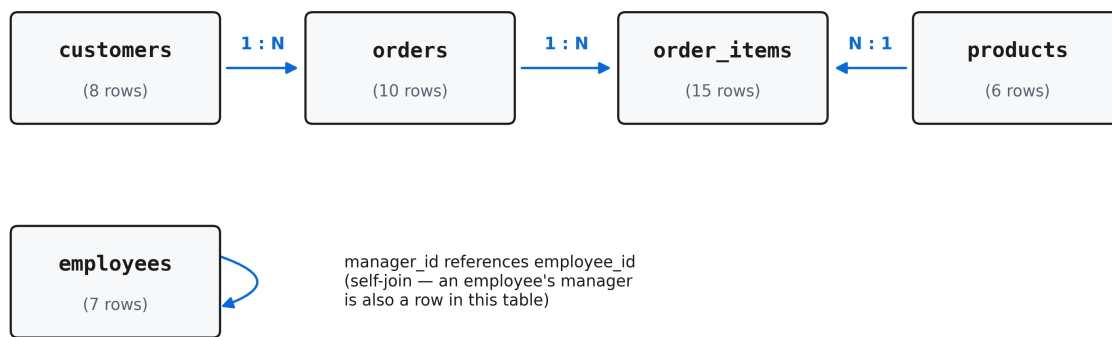


Figure 1: Entity relationship diagram of the BrightCart database

Foreign Key	In Table	Column	References	Meaning
Customer → Orders	orders	customer_id	customers.customer_id	Each order belongs to one customer; one customer can have many orders
Order → Line Items	order_items	order_id	orders.order_id	Each line item belongs to one order; one order can have many line items
Product → Line Items	order_items	product_id	products.product_id	Each line item refers to one product; one product can appear in many line items
Manager → Employee	employees	manager_id	employees.employee_id	Each employee may report to one manager, who is also an employee

The `order_items` table sits at the center — it connects orders to products and stores the actual quantity and price for each purchase. This is a common pattern in retail databases.

Two important things to notice:

1. **products.unit_price** is the current catalog price. But prices change over time. The price a customer actually paid is stored in **order_items.sale_price**. These two numbers can be different.
2. **orders.status** tells you whether an order was actually fulfilled. Not every order in the database represents real revenue — some were cancelled, returned, or are still pending.

4 — Setting Up the Data in qStudio

Open qStudio, connect to DuckDB, and run the following SQL blocks one at a time. This creates the five tables and fills them with realistic data.

4.1 — Create the Tables

```
CREATE OR REPLACE TABLE customers (  
    customer_id INTEGER PRIMARY KEY,  
    first_name VARCHAR,  
    last_name VARCHAR,  
    city VARCHAR,  
    state VARCHAR,  
    join_date DATE  
);  
  
CREATE OR REPLACE TABLE products (  
    product_id INTEGER PRIMARY KEY,  
    product_name VARCHAR,  
    category VARCHAR,  
    unit_price DECIMAL(10,2),  
    discontinued BOOLEAN  
);  
  
CREATE OR REPLACE TABLE orders (  
    order_id INTEGER PRIMARY KEY,  
    customer_id INTEGER REFERENCES customers(customer_id),  
    order_date DATE,  
    status VARCHAR  
);  
  
CREATE OR REPLACE TABLE order_items (  
    item_id INTEGER PRIMARY KEY,  
    order_id INTEGER REFERENCES orders(order_id),  
    product_id INTEGER REFERENCES products(product_id),  
    quantity INTEGER,  
    sale_price DECIMAL(10,2)  
);  
  
CREATE OR REPLACE TABLE employees (  
    employee_id INTEGER PRIMARY KEY,  
    first_name VARCHAR,  
    last_name VARCHAR,  
    department VARCHAR,  
    hire_date DATE,  
    salary DECIMAL(10,2),  
    manager_id INTEGER REFERENCES employees(employee_id)  
);
```

4.2 — Insert the Data

-- 8 customers (two of them have never placed an order)

```
INSERT INTO customers VALUES
  (1, 'Alice', 'Chen', 'San Jose', 'CA', '2023-01-15'),
  (2, 'Bob', 'Martinez', 'San Francisco', 'CA', '2023-03-22'),
  (3, 'Carol', 'Johnson', 'Seattle', 'WA', '2023-06-10'),
  (4, 'David', 'Kim', 'Portland', 'OR', '2023-09-01'),
  (5, 'Emma', 'Williams', 'San Jose', 'CA', '2024-01-05'),
  (6, 'Frank', 'Brown', 'Denver', 'CO', '2024-02-14'),
  (7, 'Grace', 'Lee', 'Austin', 'TX', '2024-04-20'),
  (8, 'Henry', 'Davis', 'Phoenix', 'AZ', '2024-06-30');
```

-- 6 products (the Mouse and Desk Mat have had price increases)

-- Mouse was \$24.99, now \$29.99

-- Desk Mat was \$39.99, now \$45.99

```
INSERT INTO products VALUES
  (101, 'Wireless Mouse', 'Electronics', 29.99, false),
  (102, 'USB-C Hub', 'Electronics', 49.99, false),
  (103, 'Notebook 3-Pack', 'Office', 12.99, false),
  (104, 'Standing Desk Mat', 'Furniture', 45.99, false),
  (105, 'Webcam HD', 'Electronics', 79.99, false),
  (106, 'Paper Shredder', 'Office', 89.99, true);
```

-- 10 orders (not all were successfully delivered)

```
INSERT INTO orders VALUES
  (1001, 1, '2024-10-05', 'Delivered'),
  (1002, 1, '2024-11-12', 'Delivered'),
  (1003, 2, '2024-10-18', 'Delivered'),
  (1004, 2, '2024-11-25', 'Cancelled'),
  (1005, 3, '2024-10-02', 'Delivered'),
  (1006, 3, '2024-12-01', 'Returned'),
  (1007, 4, '2024-10-20', 'Delivered'),
  (1008, 5, '2024-11-01', 'Delivered'),
  (1009, 5, '2024-12-10', 'Pending'),
  (1010, 6, '2024-10-10', 'Delivered');
```

-- 15 line items (sale_price = what customer actually paid)

```
INSERT INTO order_items VALUES
  (1, 1001, 101, 2, 24.99),
  (2, 1001, 103, 1, 12.99),
  (3, 1002, 102, 1, 49.99),
  (4, 1003, 105, 1, 79.99),
  (5, 1003, 101, 1, 24.99),
  (6, 1004, 104, 2, 39.99),
  (7, 1005, 102, 1, 49.99),
  (8, 1005, 103, 3, 12.99),
  (9, 1006, 105, 1, 79.99),
```

```
(10, 1007, 101, 3, 24.99),
(11, 1007, 104, 1, 39.99),
(12, 1008, 102, 2, 49.99),
(13, 1008, 103, 1, 12.99),
(14, 1009, 105, 1, 79.99),
(15, 1010, 101, 1, 24.99);
```

-- 7 employees

-- -----
-- 2 managers first
-- -----

INSERT INTO employees **VALUES**

```
(1, 'Sarah', 'Wilson', 'Sales', '2020-03-15', 75000, NULL),
(4, 'James', 'Thomas', 'Engineering', '2022-01-20', 95000, NULL);
```

-- -----
-- 5 others: rest of employees
-- 2 have no department assigned yet
-- -----

INSERT INTO employees **VALUES**

```
(2, 'Mike', 'Taylor', 'Sales', '2021-06-01', 62000, 1),
(3, 'Lisa', 'Anderson', 'Marketing', '2021-09-10', 68000, 1),
(5, 'Nina', 'Patel', 'Engineering', '2022-08-15', 85000, 4),
(6, 'Carlos', 'Rivera', NULL, '2024-07-01', 55000, 1),
(7, 'Amy', 'Zhang', NULL, '2024-07-15', 58000, 4);
```

4.3 — Verify the Data

Run this quick check to confirm everything loaded:

```
SELECT 'customers' AS table_name, COUNT(*) AS rows FROM customers
UNION ALL
SELECT 'products', COUNT(*) FROM products
UNION ALL
SELECT 'orders', COUNT(*) FROM orders
UNION ALL
SELECT 'order_items', COUNT(*) FROM order_items
UNION ALL
SELECT 'employees', COUNT(*) FROM employees;
```

You should see:

table_name varchar	rows int64
customers	8

products	6
orders	10
order_items	15
employees	7

5 — Example 1: The Missing Customers — AI Drops Data You Need

The Business Question

Your manager asks: “How many orders has each customer placed? I want to see all customers, including those who haven’t ordered yet.”

What You Asked AI

You open a chatbot and type:

“Using DuckDB, write SQL to show how many orders each customer has placed. I want to see all customers, including those with no orders. Tables: customers and orders, linked by customer_id.”

The prompt is clear — it even says “including those with no orders.” Let’s see what the AI produces.

What AI Generated

```
SELECT  c.first_name,
        c.last_name,
        COUNT(o.order_id) AS total_orders
FROM    customers c
        JOIN orders o
          ON c.customer_id = o.customer_id
GROUP BY c.first_name, c.last_name
ORDER BY total_orders DESC;
```

The AI's Answer

first_name varchar	last_name varchar	total_orders int64
Alice	Chen	2
Carol	Johnson	2
Bob	Martinez	2
Emma	Williams	2
David	Kim	1
Frank	Brown	1

The query runs perfectly. No errors. Six customers, each with at least one order. Looks good, right?

What's Wrong

BrightCart has 8 customers, not 6. Where are Grace Lee and Henry Davis?

The AI used a JOIN (also called an INNER JOIN), which only returns customers that have a matching row in the orders table. Grace and Henry have never placed an order, so they have no match — and they silently disappear from the results.

The manager explicitly asked for all customers, including those who haven't ordered. The AI ignored that requirement.

The Correct SQL

```
SELECT  c.first_name,
        c.last_name,
        COUNT(o.order_id) AS total_orders
FROM    customers c
        LEFT JOIN orders o
          ON c.customer_id = o.customer_id
GROUP BY c.first_name, c.last_name
ORDER BY total_orders DESC;
```

The Correct Answer

first_name varchar	last_name varchar	total_orders int64
Alice	Chen	2
Emma	Williams	2
Carol	Johnson	2
Bob	Martinez	2
David	Kim	1
Frank	Brown	1
Grace	Lee	0
Henry	Davis	0

Now we see all 8 customers. Grace and Henry show 0 orders — exactly the information the manager needed.

The Business Impact

If you used the AI's answer, you would report: "Our least active customer has 1 order. Everyone is buying." In reality, 25% of your customers have never purchased anything. That's critical information for a re-engagement campaign. The AI's answer hid it completely.

The Lesson: JOIN and LEFT JOIN are not the same thing. A single wrong word can make entire rows of data disappear from your results — and the query will still run without any error.

6 — Example 2: The Wrong Revenue — AI Uses the Wrong Price

The Business Question

Your manager asks: “What was our total revenue from all orders?”

What You Asked AI

“Write a DuckDB query to calculate total revenue from all orders. Tables: customers, products, orders, order_items.”

Notice what’s missing from the prompt: you didn’t specify which price column to use — because you don’t know there are two. The AI has to guess, and it guesses wrong.

What AI Generated

```
SELECT SUM(oi.quantity * p.unit_price) AS total_revenue
FROM   order_items oi
       JOIN products p
       ON oi.product_id = p.product_id;
```

The AI’s Answer

total_revenue
852.78

The query runs. One clean number. You report \$852.78 to your manager.

What’s Wrong

The AI calculated revenue using **products.unit_price** — the current catalog price. But two products have had price increases since these orders were placed:

Product	PriceWhenSold	CurrentPrice
Wireless Mouse	\$24.99	\$29.99
Standing Desk Mat	\$39.99	\$45.99

The price customers actually paid is stored in **order_items.sale_price**. The AI joined to the wrong table and used the wrong price column.

The Correct SQL

```
SELECT SUM(quantity * sale_price) AS total_revenue
FROM order_items;
```

The Correct Answer

total_revenue
799.78

The real revenue is \$799.78, not \$852.78. The AI's answer was \$53.00 too high — a 6.6% overstatement.

The Business Impact

A 6.6% revenue overstatement in a report might seem small on 10 orders. On a company doing \$10 million in annual sales, the same mistake overstates revenue by **\$660,000**. That's the kind of error that leads to wrong financial forecasts, incorrect tax filings, and very difficult conversations with auditors.

The Lesson: Databases often store the same type of information (like “price”) in multiple places. The current price and the historical price are different numbers. AI doesn't understand your business well enough to know which one to use. You do — but only if you understand the schema and the SQL.

7 — Example 3: The Inflated Numbers — AI Counts Cancelled Orders

The Business Question

Your manager asks: “How many orders did we successfully deliver in Q4 2024?”

What You Asked AI

“How many orders were delivered in Q4 2024? Table: orders.”

You said “delivered,” but you didn't tell the AI that the table has a status column that needs to be filtered. The AI treats “delivered” as a time-range description, not as a SQL filter condition.

What AI Generated

```
SELECT COUNT(*) AS orders_delivered
FROM orders
WHERE order_date BETWEEN '2024-10-01' AND '2024-12-31';
```

The AI's Answer

orders_delivered

10

Ten orders delivered in Q4. You put “10” in the quarterly report.

What's Wrong

The AI counted all orders in that date range, regardless of status. But not all orders were delivered:

order_id	status	ShouldCount?
1001	Delivered	Yes
1002	Delivered	Yes
1003	Delivered	Yes
1004	Cancelled	No
1005	Delivered	Yes
1006	Returned	No
1007	Delivered	Yes
1008	Delivered	Yes
1009	Pending	No
1010	Delivered	Yes

Three orders were not successfully delivered: one was cancelled, one was returned, and one is still pending.

The Correct SQL

```
SELECT COUNT(*) AS orders_delivered
FROM orders
WHERE order_date BETWEEN '2024-10-01' AND '2024-12-31'
      AND status = 'Delivered';
```

The Correct Answer

orders_delivered

7

The real answer is 7, not 10. The AI inflated the count by 43%.

The Business Impact

If you report 10 deliveries when only 7 actually shipped, your fulfillment metrics look artificially strong. The operations team thinks everything is fine. Meanwhile, 30% of orders are failing — a serious problem that nobody is investigating because the report said 10.

The Lesson: AI doesn't understand your business rules. It doesn't know that "delivered" means `status = 'Delivered'`, not just "exists in the orders table." Business logic lives in the data, and only someone who understands both SQL and the domain can write the correct filter.

8 — Example 4: The NULL Trap — AI Forgets About Missing Data

The Business Question

Your manager asks: "List all employees who are NOT in the Sales department."

What AI Generated

```
SELECT first_name,  
       last_name,  
       department  
FROM   employees  
WHERE  department != 'Sales';
```

The AI's Answer

first_name	last_name	department
Lisa	Anderson	Marketing
James	Thomas	Engineering
Nina	Patel	Engineering

Three employees. The rest must all be in Sales, right?

What's Wrong

BrightCart has 7 employees. Two are in Sales (Sarah and Mike). That leaves 5 who are not in Sales. But the AI only found 3.

Where are Carlos Rivera and Amy Zhang? They are recent hires whose department has not been assigned yet — their department column is NULL (empty/unknown).

In SQL, NULL is special. It does not equal anything, and it does not not-equal anything either. The comparison `NULL != 'Sales'` does not return TRUE — it returns NULL, which SQL treats as "unknown," and the row is excluded.

This is one of the most common mistakes in SQL, and AI makes it frequently.

The Correct SQL

```
SELECT first_name,  
       last_name,  
       department  
FROM   employees  
WHERE  department != 'Sales'  
OR     department IS NULL;
```

The Correct Answer

first_name	last_name	department
Lisa	Anderson	Marketing
James	Thomas	Engineering
Nina	Patel	Engineering
Carlos	Rivera	NULL
Amy	Zhang	NULL

Now we see all 5 employees who are not in Sales, including the two unassigned new hires.

The Business Impact

If you used the AI's answer to send a company-wide announcement to "everyone outside Sales," Carlos and Amy would never receive it. If you used it to calculate headcount for budget planning, your non-Sales departments would appear to have 3 people instead of 5 — understaffed by 40%.

The Lesson: NULL is not a value — it means "unknown." Standard comparisons (=, !=, >, <) do not work with NULL. You must use IS NULL or IS NOT NULL. AI frequently forgets this rule, and the query runs without any error or warning.

9 — Example 5: The Wrong Average — AI Confuses Rows with Orders

The Business Question

Your manager asks: "What is our average order value for delivered orders?"

What You Asked AI

"Calculate the average order value for delivered orders. Tables: orders and order_items, linked by order_id."

"Average order value" sounds clear in English, but it is ambiguous in SQL: do you mean the average of each line item's value, or the average of each order's total? The AI picks the simpler interpretation — and gets the wrong number.

What AI Generated

```
SELECT ROUND(AVG(quantity * sale_price), 2) AS avg_order_value
FROM   order_items oi
      JOIN orders o
      ON oi.order_id = o.order_id
WHERE  o.status = 'Delivered';
```

The AI's Answer

avg_order_value
46.65

\$46.65 per order. You put this number in the financial summary.

What's Wrong

The AI averaged the individual line items, not the order totals.

Here are the delivered orders and their actual totals:

order_id	Line Items	Order Total
1001	(2 × \$24.99) + (1 × \$12.99)	\$62.97
1002	(1 × \$49.99)	\$49.99
1003	(1 × \$79.99) + (1 × \$24.99)	\$104.98
1005	(1 × \$49.99) + (3 × \$12.99)	\$88.96
1007	(3 × \$24.99) + (1 × \$39.99)	\$114.96
1008	(2 × \$49.99) + (1 × \$12.99)	\$112.97
1010	(1 × \$24.99)	\$24.99

The average of these 7 order totals is: $(\$62.97 + \$49.99 + \$104.98 + \$88.96 + \$114.96 + \$112.97 + \$24.99) \div 7 = \79.97

The AI's answer of \$46.65 is 42% too low because it averaged 12 line items instead of 7 order totals. An order with 3 line items got 3 times the weight of an order with 1 line item.

The Correct SQL

```
SELECT ROUND(AVG(order_total), 2) AS avg_order_value
FROM (
  SELECT o.order_id,
         SUM(oi.quantity * oi.sale_price) AS order_total
  FROM   orders o
        JOIN order_items oi
        ON o.order_id = oi.order_id)
```

```

WHERE    o.status = 'Delivered'
GROUP BY o.order_id
) AS order_totals;

```

The Correct Answer

avg_order_value
79.97

The real average order value is \$79.97, not \$46.65.

The Business Impact

Average order value is a key metric for pricing strategy, marketing spend, and revenue forecasting. If you report \$47 instead of \$80, management might conclude that customers aren't spending enough and launch unnecessary discounts — cutting into margins to solve a problem that doesn't exist.

The Lesson: “Average” depends on what you're averaging. Are you averaging line items or orders? AI often gets the “grain” wrong — it computes the average at the wrong level of detail. The query runs fine. The number looks reasonable. But it's the wrong number.

10 — Example 6: The Broken Ranking — AI Ignores Ties

The Business Question

Your manager asks: “Rank our products by total units sold from delivered orders. If two products sold the same quantity, they must share the same rank.”

What AI Generated

```

SELECT    p.product_name,
          SUM(oi.quantity) AS units_sold,
          ROW_NUMBER() OVER (ORDER BY SUM(oi.quantity) DESC) AS sales_rank
FROM      order_items oi
          JOIN products p
            ON oi.product_id = p.product_id
          JOIN orders o
            ON oi.order_id = o.order_id
WHERE     o.status = 'Delivered'
GROUP BY  p.product_name;

```

The AI's Answer

product_name	units_sold	sales_rank
Wireless Mouse	7	1
Notebook 3-Pack	5	2
USB-C Hub	4	3
Standing Desk Mat	1	4
Webcam HD	1	5

Five products, each with a unique rank. Looks clean, right?

What's Wrong

Look at the bottom two products: the Standing Desk Mat and the Webcam HD both sold exactly 1 unit. They are tied. The manager explicitly asked for tied products to share the same rank.

But the AI used **ROW_NUMBER()**, which assigns a unique sequential number to every row — it never produces ties. The Desk Mat got rank 4 and the Webcam got rank 5 purely based on arbitrary internal ordering, not on actual performance.

The correct function is **RANK()**, which gives tied rows the same rank number.

The Correct SQL

```
SELECT  p.product_name,
        SUM(oi.quantity) AS units_sold,
        RANK() OVER (ORDER BY SUM(oi.quantity) DESC) AS sales_rank
FROM    order_items oi
        JOIN products p
          ON oi.product_id = p.product_id
        JOIN orders o
          ON oi.order_id = o.order_id
WHERE   o.status = 'Delivered'
GROUP BY p.product_name;
```

The Correct Answer

product_name	units_sold	sales_rank
Wireless Mouse	7	1
Notebook 3-Pack	5	2
USB-C Hub	4	3
Standing Desk Mat	1	4
Webcam HD	1	4

Now both products that sold 1 unit share rank 4 — exactly what the manager asked for.

The Business Impact

Imagine the ranking is used to decide which products to discontinue: “Cut the lowest-ranked product.” With the AI’s answer, only the Webcam (rank 5) gets flagged. The Desk Mat (rank 4) looks like it performed better — even though they had identical sales. You might discontinue one product while keeping another that performed exactly the same. Fair decisions require accurate rankings.

The Lesson: ROW_NUMBER(), RANK(), and DENSE_RANK() are three different functions that look similar but behave differently with ties. AI frequently picks the wrong one. If your business question involves ties, you need to know the difference — and verify which function the AI chose.

11 — Example 7: The LIMIT Illusion — AI Can’t Do “Per Customer”

The Business Question

Your manager asks: “For each customer who has placed an order, show me their single most recent order with the order date and status.”

What AI Generated

```
SELECT  c.first_name,
        c.last_name,
        o.order_id,
        o.order_date,
        o.status
FROM    customers c
JOIN    orders o
      ON c.customer_id = o.customer_id
ORDER BY o.order_date DESC
LIMIT   6;
```

The AI’s Answer

first_name	last_name	order_id	order_date	status
Emma	Williams	1009	2024-12-10	Pending
Carol	Johnson	1006	2024-12-01	Returned
Bob	Martinez	1004	2024-11-25	Cancelled
Alice	Chen	1002	2024-11-12	Delivered
Emma	Williams	1008	2024-11-01	Delivered
David	Kim	1007	2024-10-20	Delivered

Six rows. Six customers have orders, so one row per customer, right?

What's Wrong

Look carefully: Emma Williams appears twice (orders 1009 and 1008). And Frank Brown is completely missing — his most recent order (1010, October 10) didn't make the global top-6 cutoff.

The AI used `ORDER BY ... LIMIT 6`, which returns the 6 most recent orders across all customers. It does not return the most recent order for each customer. `LIMIT` is a global cap — it has no concept of “per group.”

To get one row per customer, you need a window function that partitions the data by customer and picks the top row within each partition.

The Correct SQL

```
SELECT first_name,
        last_name,
        order_id,
        order_date,
        status
FROM (
    SELECT c.first_name,
            c.last_name,
            o.order_id,
            o.order_date,
            o.status,
            ROW_NUMBER() OVER (
                PARTITION BY c.customer_id
                ORDER BY o.order_date DESC
            ) AS rn
    FROM customers c
    JOIN orders o
        ON c.customer_id = o.customer_id
) AS ranked
WHERE rn = 1
ORDER BY order_date DESC;
```

The Correct Answer

first_name	last_name	order_id	order_date	status
Emma	Williams	1009	2024-12-10	Pending
Carol	Johnson	1006	2024-12-01	Returned
Bob	Martinez	1004	2024-11-25	Cancelled
Alice	Chen	1002	2024-11-12	Delivered
David	Kim	1007	2024-10-20	Delivered
Frank	Brown	1010	2024-10-10	Delivered

Now each customer appears exactly once with their most recent order. Frank is back.

The Business Impact

If you used the AI's answer to contact each customer about their most recent experience, Emma would receive two messages while Frank would receive none. You would also miss that Frank's last order was back in October — he may be at risk of churning and could benefit from a follow-up. Meanwhile, the duplicated effort on Emma wastes resources and looks unprofessional.

The Lesson: LIMIT restricts the total number of rows returned — it cannot select “the top N per group.” Whenever you need the top row (or top N rows) within each group, you need a window function like ROW_NUMBER() OVER (PARTITION BY ...). AI frequently reaches for LIMIT when a window function is required.

Part II — When AI Gets It Right

The examples above might make it seem like AI always fails. It doesn't. AI can produce correct SQL — and when it does, it saves time. But here's the catch: you can only benefit from AI's correct answers if you have the knowledge to recognize them as correct.

12 — Example 8: Top 3 Customers — Can You Verify It?

The Business Question

Your manager asks: “Who are our top 3 customers by total spending on delivered orders?”

What AI Generated

```
SELECT  c.first_name,
        c.last_name,
        SUM(oi.quantity * oi.sale_price) AS total_spent
FROM    customers c
        JOIN orders o
          ON c.customer_id = o.customer_id
        JOIN order_items oi
          ON o.order_id = oi.order_id
WHERE   o.status = 'Delivered'
GROUP BY c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT  3;
```

The AI's Answer

first_name	last_name	total_spent
David	Kim	114.96

first_name	last_name	total_spent
Emma	Williams	112.97
Alice	Chen	112.96

Is This Correct?

Yes. This query is correct. It joins the right tables, filters by delivered status, uses sale_price (not unit_price), groups by customer, sorts descending, and limits to the top 3.

So Why Do You Still Need to Know SQL?

Because how would you know it's correct if you didn't understand the SQL?

To verify this answer, you need to understand:

Concept	Why You Need It
JOIN (three tables)	Are customers, orders, and items connected correctly?
WHERE status='Delivered'	Did the AI exclude non-delivered orders? (Example 3 showed it doesn't always do this)
SUM(oi.quantity*oi.sale_price)	Did it use the right price column? (Example 2 showed it sometimes picks the wrong one)
GROUPBY	Is it grouping by customer, not by order or by line item? (Example 5 showed the grain matters)
ORDERBY...DESC	Is it sorting highest-to-lowest?
LIMIT3	Is it showing the top 3, not the bottom 3?

If you cannot read the SQL, you cannot answer any of these questions. You are forced to blindly trust that the AI got it right — even though the seven examples above proved it often doesn't.

The Lesson: Even when AI writes correct SQL, your job is to verify it. Verification requires understanding. Understanding requires learning. There is no shortcut.

13 — Example 9: Revenue by Category — Can You Prove It?

The Business Question

Your manager asks: "Show me total revenue by product category for delivered orders only. Include the number of orders and total units sold for each category. Use the actual price the customer paid."

What AI Generated

```
SELECT p.category,
       COUNT(DISTINCT o.order_id) AS num_orders,
```

```

SUM(oi.quantity) AS total_units,
SUM(oi.quantity * oi.sale_price) AS total_revenue
FROM order_items oi
JOIN orders o
ON oi.order_id = o.order_id
JOIN products p
ON oi.product_id = p.product_id
WHERE o.status = 'Delivered'
GROUP BY p.category
ORDER BY total_revenue DESC;

```

The AI's Answer

category	num_orders	total_units	total_revenue
Electronics	7	12	454.88
Office	3	5	64.95
Furniture	1	1	39.99

Is This Correct?

Yes. This query is correct. But don't take our word for it — prove it yourself.

Student Challenge

Your task: This query is correct, but can you explain why? Examine the SQL carefully and identify at least five things the AI got right. For each one, reference which earlier example showed AI getting that same thing wrong.

Here is a checklist to guide your analysis:

What to Check	Which Example Showed This Going Wrong?
Does the JOIN type matter here? Could LEFT JOIN change the result?	Example 1 (The Missing Customers)
Which price column does the query use — sale_price or unit_price?	Example 2 (The Wrong Revenue)
Does the query filter by order status?	Example 3 (The Inflated Numbers)
Could NULL values in any column affect the result?	Example 4 (The NULL Trap)
Is the GROUPBY at the right level of detail? Is COUNT(DISTINCT...) necessary?	Example 5 (The Wrong Average)
Does the query need RANK() or ROW_NUMBER()? Why or why not?	Example 6 (The Broken Ranking)
Does LIMIT appear? Should it?	Example 7 (The LIMIT Illusion)

Bonus Verification

If you want to double-check the numbers, you can verify the Electronics total manually. The delivered orders containing electronics products are:

order_id	product	qty	sale_price	line_total
1001	Wireless Mouse	2	\$24.99	\$49.98
1002	USB-C Hub	1	\$49.99	\$49.99
1003	Webcam HD	1	\$79.99	\$79.99
1003	Wireless Mouse	1	\$24.99	\$24.99
1005	USB-C Hub	1	\$49.99	\$49.99
1007	Wireless Mouse	3	\$24.99	\$74.97
1008	USB-C Hub	2	\$49.99	\$99.98
1010	Wireless Mouse	1	\$24.99	\$24.99

Electronics total:

$\$49.98 + \$49.99 + \$79.99 + \$24.99 + \$49.99 + \$74.97 + \$99.98 + \$24.99 = \$454.88 \checkmark$

All three categories combined:

$\$454.88 + \$64.95 + \$39.99 = \559.82

— which matches the total delivered revenue from our earlier examples.

The Lesson: The best way to trust AI output is to understand it deeply enough to verify it yourself. When you can read the SQL, check the logic, and confirm the numbers, AI becomes a powerful accelerator. Without that understanding, it's just a black box.

14 — Example 10: Department Salary Report — AI Remembers NULL

The Business Question

Your manager asks: “For each department that has employees assigned, show the headcount, average salary, lowest salary, and highest salary.”

What AI Generated

```
SELECT    department,
          COUNT(*)           AS num_employees,
          ROUND(AVG(salary), 0) AS avg_salary,
          MIN(salary)        AS min_salary,
          MAX(salary)        AS max_salary
FROM      employees
WHERE     department IS NOT NULL
GROUP BY  department
ORDER BY  avg_salary DESC;
```

The AI's Answer

department	num_employees	avg_salary	min_salary	max_salary
Engineering	2	90000	85000	95000
Sales	2	68500	62000	75000
Marketing	1	68000	68000	68000

Is This Correct?

Yes. Let's verify against the raw data:

employee	department	salary
Sarah Wilson	Sales	\$75,000
Mike Taylor	Sales	\$62,000
Lisa Anderson	Marketing	\$68,000
James Thomas	Engineering	\$95,000
Nina Patel	Engineering	\$85,000
Carlos Rivera	NULL	\$55,000
Amy Zhang	NULL	\$58,000

Engineering: 2 employees, avg $(\$95,000 + \$85,000) \div 2 = \$90,000$ ✓

Sales: 2 employees, avg $(\$75,000 + \$62,000) \div 2 = \$68,500$ ✓

Marketing: 1 employee, avg \$68,000 ✓

Carlos and Amy are correctly excluded by `WHERE department IS NOT NULL`.

Why This Matters

Remember Example 4? There the AI forgot about NULL and silently dropped employees from the result. Here, the AI handled NULL correctly — it used `IS NOT NULL` to exclude unassigned employees, which is exactly what the manager requested.

But you could only confirm this by reading the SQL. If you didn't understand the `WHERE department IS NOT NULL` clause, you might not realize that two employees were intentionally excluded — or worse, you might not notice they were missing at all.

Think About It

Question for you: What would happen if we removed the `WHERE department IS NOT NULL` line? Would the query still run? What would appear in the department column for Carlos and Amy? Would the averages change?

Try it in qStudio and see for yourself.

The Lesson: When AI gets NULL handling right, it's because the pattern happened to match its training data — not because it understood your business rules. Your job is to confirm that the NULL handling matches what the manager actually wanted.

15 — Example 11: Full Product Catalog with Sales — AI Uses a Subquery

The Business Question

Your manager asks: “Show me every product in our catalog — including products that have never been sold — along with their total units sold and total revenue from delivered orders. Products with no sales should show zero.”

This is a challenging query because it combines two separate ideas: “all products” (even unsold ones) and “only delivered revenue.”

What AI Generated

```
SELECT  p.product_name,
        p.category,
        p.unit_price AS current_price,
        COALESCE(d.total_units, 0) AS total_units_sold,
        COALESCE(d.total_revenue, 0) AS total_revenue
FROM    products p
        LEFT JOIN (
            SELECT  oi.product_id,
                    SUM(oi.quantity) AS total_units,
                    SUM(oi.quantity * oi.sale_price) AS total_revenue
            FROM    order_items oi
                    JOIN orders o
                      ON oi.order_id = o.order_id
            WHERE   o.status = 'Delivered'
            GROUP BY oi.product_id
        ) d ON p.product_id = d.product_id
ORDER BY total_revenue DESC;
```

The AI's Answer

product_name	category	current_price	total_units_sold	total_revenue
USB-C Hub	Electronics	49.99	4	199.96
Wireless Mouse	Electronics	29.99	7	174.93
Webcam HD	Electronics	79.99	1	79.99
Notebook 3-Pack	Office	12.99	5	64.95
Standing Desk Mat	Furniture	45.99	1	39.99
Paper Shredder	Office	89.99	0	0.00

Is This Correct?

Yes. This is a sophisticated query, and the AI nailed it. Let's break down why it works:

Technique	Why It's Here	Which Lesson Applies
LEFTJOIN to the subquery	Ensures all products appear, even the Paper Shredder with zero sales	Example 1 (Missing Customers)
Subquery uses sale_price	Revenue is based on what customers paid, not today's catalog price	Example 2 (Wrong Revenue)
Subquery filters status='Delivered'	Only counts revenue from fulfilled orders	Example 3 (Inflated Numbers)
COALESCE(d.total_units,0)	Converts NULL (from the LEFT JOIN) to 0 for unsold products	Example 4 (NULL Trap)
GROUPBYoi.product_id in subquery	Aggregates at the product level before joining back	Example 5 (Wrong Average)

Verification

The six product totals add up:

$\$199.96 + \$174.93 + \$79.99 + \$64.95 + \$39.99 + \$0.00 = \$559.82$

— which matches the total delivered revenue we verified in earlier examples. ✓

Notice that the Paper Shredder (product 106, discontinued) is included with zero sales. It was never ordered, so the LEFT JOIN produces NULL values, and COALESCE converts them to 0. This is exactly what the manager asked for.

Think About It

Question for you: The current_price column shows today's catalog price (\$29.99 for the Wireless Mouse), but the total_revenue column is based on the historical sale_price (\$24.99 per unit). That means $\$29.99 \times 7 \text{ units} = \209.93 , but the actual revenue is \$174.93. Why is this difference a feature, not a bug? What would go wrong if the query used current_price to calculate revenue?

(Hint: Review Example 2.)

The Lesson: When AI produces a correct complex query, every piece of it — the JOIN type, the price column, the status filter, the NULL handling, the aggregation level — represents a decision that could have gone wrong. You saw seven examples where it did. The ability to read this query and confirm that every decision was the right one is exactly why you need to learn SQL.

Closing

16 — The Verdict

AI is a powerful tool. It can save time, suggest approaches you haven't considered, and help you write SQL faster — once you already understand SQL.

But AI is not a substitute for understanding. Here is what went wrong in the seven examples from Part I:

Example	AI's Answer	Correct Answer	Error
1. Customer count	6 customers	8 customers	2 customers missing (25%)
2. Total revenue	\$852.78	\$799.78	\$53.00 overstated (6.6%)
3. Delivered orders	10 orders	7 orders	3 extra counted (43% inflated)
4. Non-Sales employees	3 employees	5 employees	2 employees invisible (40%)
5. Average order value	\$46.65	\$79.97	\$33.32 too low (42%)
6. Product ranking	5 unique ranks	2 products share rank 4	Tied products given different ranks
7. Most recent order	Emma appears twice	Each customer once	1 customer duplicated, 1 missing

Every one of these queries ran without a single error message. Every one returned a clean, professional-looking table. Every one was wrong.

In a real company, these mistakes lead to:

- Wrong financial reports sent to executives
- Marketing campaigns that target the wrong customers
- Inventory decisions based on inflated demand
- Budget allocations based on missing headcount
- Strategic plans built on incorrect metrics
- Unfair product or employee evaluations
- Duplicated outreach and missed at-risk customers

You would not trust a financial report you cannot read. Do not trust a SQL query you cannot read either.

17 — Summary of Lessons

Part I — When AI Gets It Wrong

#	Trap	What AI Did Wrong	What You Need to Know
1	Wrong JOIN type	Used JOIN instead of LEFTJOIN — dropped customers with no orders	The difference between JOIN and LEFTJOIN
2	Wrong price column	Used the current catalog price instead of the historical sale price	How to read a schema and pick the right column

#	Trap	What AI Did Wrong	What You Need to Know
3	Missing business filter	Counted all orders, not just delivered ones	How to apply business rules with WHERE
4	NULL behavior	Used != which silently skips NULL values	How NULL works in SQL comparisons
5	Wrong aggregation grain	Averaged line items instead of order totals	How GROUPBY and subqueries control the level of detail
6	Wrong ranking function	Used ROW_NUMBER() instead of RANK() — ignored ties	The difference between ROW_NUMBER(), RANK(), and DENSE_RANK()
7	LIMIT instead of window	Used LIMIT for a “per group” question — duplicated some, missed others	How ROW_NUMBER()OVER(PARTITIONBY...) solves per-group problems

Part II — When AI Gets It Right

#	What AI Got Right	Why You Still Need SQL
8	Top 3 customers — correct JOINS, filter, price column, grouping	You must verify each decision the AI made; blind trust is not an option
9	Revenue by category — correct across all dimensions	Systematic verification ties back to every lesson in Part I
10	Department salaries — correctly handled NULL exclusion	Confirming NULL logic requires understanding ISNOTNULL
11	Product catalog with sales — LEFT JOIN, subquery, COALESCE, status filter	Every technique from Part I appears in one query; you must verify them all

The Bottom Line

Learn SQL first. Then use AI to write it faster. Never the other way around.

18 — How to Prompt AI Better

This document proved that blind trust in AI-generated SQL is dangerous. But the answer is not to avoid AI — it is to use it properly. Once you understand SQL, AI becomes a powerful accelerator. Here are seven rules for getting better results.

Rule 1 — Include the Schema

Don’t say “write a query for our orders database.” Give the AI the actual CREATE TABLE statements. The more context it has about your columns, types, and relationships, the better its choices will be.

Weak prompt: > “Write SQL to find total revenue by customer.”

Strong prompt: > “Given these tables: [paste your CREATE TABLE statements]. Write SQL > to find total revenue per customer. Revenue should use > order_items.sale_price (the price

at time of sale), not `> products.unit_price` (the current catalog price). Only include orders `> where status = 'Delivered'.`

The strong prompt eliminates Examples 2 and 3 before the AI even starts.

Rule 2 — State the Business Rules Explicitly

AI does not know your business. If “revenue” only counts delivered orders, say so. If employees with NULL departments should be excluded, say so. If tied products must share the same rank, say so. Every unstated assumption is an opportunity for a mistake.

Think of it this way: the seven failures in Part I all came from business rules the AI had to guess. The fewer guesses, the fewer errors.

Rule 3 — Specify Edge Cases

Mention NULLs, ties, cancelled orders, customers with no purchases — whatever applies to your data. Examples 1, 4, 6, and 7 all involved edge cases that the AI ignored.

A useful prompt addition: `> “Note: some employees have NULL in the department column. Handle > them according to [your requirement].”`

Rule 4 — Name the SQL Dialect

Don’t just say “SQL” — say “DuckDB SQL.” Left unstated, the AI defaults to whatever dialect shows up most in its training data — usually MySQL or generic ANSI SQL — and that can produce queries that fail outright, or worse, run and quietly behave differently.

Weak prompt: `> “Write SQL to get the top 3 products per category.”`

Strong prompt: `> “Write DuckDB SQL to get the top 3 products per category.”`

A few real examples of what goes wrong without it:

MySQL-flavored answer	What happens in DuckDB
<code>'column_name'</code> (backticks)	Syntax error — DuckDB expects double quotes
<code>LIMIT10,3</code> (offset, count)	Syntax error — DuckDB wants <code>LIMIT3OFFSET10</code>
<code>DATE_FORMAT(order_date,'%Y-%m')</code>	Function doesn’t exist — DuckDB uses <code>strftime(order_date,'%Y-%m')</code>

Naming the dialect also works in your favor, not just against errors: it unlocks DuckDB-specific features — `QUALIFY`, `PIVOT/UNPIVOT`, `LIST/UNNEST` — that you’ll meet later in this course and that can make a query shorter than a generic ANSI SQL answer would be.

Rule 5 — Ask the AI to Explain Its Choices

After the AI generates SQL, ask follow-up questions:

“Why did you use JOIN instead of LEFT JOIN?”

“Why did you use ROW_NUMBER instead of RANK?”

“Which price column did you choose and why?”

“How does this query handle NULL values?”

If the AI cannot justify its decisions clearly, that is a red flag. A correct query should have a clear explanation for every choice.

Rule 6 — Verify with Known Data

Run the AI’s query on a small dataset where you already know the correct answer — like the BrightCart data in this document. If the numbers don’t match your hand calculations, you have caught a bug before it reaches a real report.

This is the same verification technique we used in Examples 8–11. It works because you built the knowledge to check the output.

Rule 7 — Read the SQL Before You Run It

Even if you trust the AI, read the query first. Use this checklist:

Check	What to Look For	Which Example Showed This Failing
JOIN type	JOIN vs LEFTJOIN — will any rows be silently dropped?	Example 1
Price column	sale_price vs unit_price — historical or current?	Example 2
Status filter	Is WHEREstatus=’Delivered’ present when needed?	Example 3
NULL handling	Does the query use ISNULL / ISNOTNULL correctly?	Example 4
Aggregation level	Is GROUPBY at the right grain (orders vs line items)?	Example 5
Ranking function	ROW_NUMBER() vs RANK() vs DENSE_RANK() — does it handle ties?	Example 6
LIMIT vs window	Is LIMIT being used where PARTITIONBY is needed?	Example 7

This checklist is exactly what you built by studying the eleven examples in this document. It turns you from a passive consumer of AI output into an active, critical reviewer.

Put It All Together: A Reusable Prompt Template

Rules 1–4 are about what you give the AI before it writes anything. Bake all four into one template, fill in the brackets, and you eliminate most of the mistakes from Part I before the AI writes a single line:

Given these tables (DuckDB SQL):
[paste your CREATE TABLE statements]

Write a DuckDB query to: [your specific question].

Business rules:

- [e.g., "revenue" means order_items.sale_price, not products.unit_price]
- [e.g., only include orders where status = 'Delivered']
- [e.g., include every customer, even those with zero orders]

Edge cases to handle:

- [e.g., some rows have NULL in [column] – include or exclude them?]
- [e.g., if two rows tie, they must share the same rank]

Rules 5–7 are about what you do after the AI answers: ask it to explain its choices, verify the numbers against data you already know, and read the query yourself before you trust it.

Try It Yourself

Go back to the weak prompt from Example 5 (Section 9): “Calculate the average order value for delivered orders. Tables: orders and order_items, linked by order_id.” That prompt produced the wrong answer — \$46.65 instead of \$79.97.

Using the template above, rewrite it as a strong prompt. Then paste your version into an AI chatbot. Does it get \$79.97 on the first try? If not, which rule did your prompt still miss?

The Real Skill

The goal of this course is not to make AI unnecessary — it is to make you the person who can tell whether AI’s answer is right or wrong.

That is a skill no AI can replace.

19 — Flagship Example: One Bad Prompt, One Good Prompt, Two Very Different Answers

Every example so far showed you a wrong query and a hand-corrected one. This example does something closer to what you’ll actually do at work: it shows the same question, asked two different ways — a weak prompt and a prompt that applies Rules 1–3 from Section 18 — and lets you watch the AI produce two completely different business answers from the exact same data.

The Business Question

Your manager asks: “Which of our customers are repeat customers — meaning they have more than one delivered order? For each one, show how many delivered orders they placed and their total spending across those delivered orders.”

This is exactly the kind of question a marketing team asks to build a loyalty program: find the best, most loyal customers and reward them. It's a completely reasonable, everyday business request.

The Weak Prompt

You're in a hurry, so you type the question into a chatbot without applying any of the rules from Section 18:

"Write SQL to find customers with more than one order, and their total spending. Tables: customers, orders, order_items."

Notice everything this leaves out: no schema, no mention of "delivered," no mention of which price column to use, and no warning that a single order can contain several line items.

What AI Generated

```
SELECT  c.first_name,
        c.last_name,
        COUNT(*) AS order_count,
        SUM(oi.quantity * p.unit_price) AS total_spent
FROM    customers c
        JOIN orders o
          ON c.customer_id = o.customer_id
        JOIN order_items oi
          ON o.order_id = oi.order_id
        JOIN products p
          ON oi.product_id = p.product_id
GROUP BY c.first_name, c.last_name
HAVING  COUNT(*) > 1
ORDER BY total_spent DESC;
```

The AI's Answer

first_name	last_name	order_count	total_spent
Bob	Martinez	3	201.96
Emma	Williams	3	192.96
Carol	Johnson	3	168.95
David	Kim	2	135.96
Alice	Chen	3	122.96

Five repeat customers, ready to load into the loyalty program. The query ran without a single error.

What's Wrong — Three Mistakes at Once

This one query repeats three mistakes from Part I simultaneously.

1. **COUNT(*)** counts line items, not orders — the fan-out bug. Joining `order_items` turns each order into one row per line item. David Kim placed exactly one order (order 1007), but that order has two line items — so **COUNT(*)** reports 2, and he is wrongly flagged as a repeat customer. This is the same wrong-grain mistake as Example 5, now showing up as a headcount instead of an average.
2. No status filter — the same mistake as Example 3. Bob’s second “order” (1004) was Cancelled. Emma’s second “order” (1009) is still Pending. Carol’s second “order” (1006) was Returned. None of these should count as evidence of repeat business or contribute revenue. The AI counted all of them anyway.
3. **products.unit_price** instead of **order_items.sale_price** — the same mistake as Example 2. Every dollar figure above uses today’s catalog price, not what the customer actually paid.

Stack those three mistakes together and the damage compounds. Once you remove the cancelled, returned, and pending orders, only Alice Chen has more than one delivered order. The other four names on this list don’t belong there at all — and this isn’t a report headed to a drawer, it’s a list of people about to receive a “thank you for your loyalty” email they didn’t earn.

The Revised Prompt

Now apply Rules 1–3 from Section 18 — schema, business rules, edge cases — before sending anything to the AI:

“Given these tables (DuckDB SQL): [paste the customers, orders, and order_items CREATE TABLE statements]. Write a DuckDB query to find repeat customers — customers with more than one delivered order. Use **COUNT(DISTINCT o.order_id)** to count orders, not **COUNT(*)**, because one order can have several line items. For revenue, use `order_items.sale_price` (the price the customer actually paid), not `products.unit_price`. Only count orders where `status = 'Delivered'` — exclude Cancelled, Returned, and Pending orders from both the order count and the total spending.”

Every sentence in this prompt closes one of the three gaps above.

What AI Generated This Time

```
SELECT  c.first_name,
        c.last_name,
        COUNT(DISTINCT o.order_id) AS order_count,
        SUM(oi.quantity * oi.sale_price) AS total_spent
FROM    customers c
        JOIN orders o
          ON c.customer_id = o.customer_id
        JOIN order_items oi
          ON o.order_id = oi.order_id
WHERE   o.status = 'Delivered'
GROUP BY c.first_name, c.last_name
```



```
HAVING    COUNT(DISTINCT o.order_id) > 1
ORDER BY total_spent DESC;
```

The Correct Answer

first_name	last_name	order_count	total_spent
Alice	Chen	2	112.96

One repeat customer, not five. And \$112.96 — not \$122.96 — is her real total. (You already verified this exact number once before, in Example 8’s Top 3 Customers list.)

Side-by-Side: The Two Prompts

	Weak Prompt	Revised Prompt
Schema included?	No	Yes (Rule 1)
“Delivered only” stated?	No	Yes (Rule 2)
Price column specified?	No	Yes (Rule 1)
Multi-line-item orders addressed?	No	Yes (Rule 3)
Repeat customers found	5	1
Top total spending shown	\$201.96	\$112.96
Correct?	No	Yes

Same manager, same question, same data — the only thing that changed was the prompt — and it changed the business decision entirely. Load the AI’s first answer into a loyalty campaign and you’d email four customers (Bob, Carol, David, Emma) who don’t qualify, while still correctly reaching the one who does.

The Lesson: A weak prompt and a strong prompt are not two versions of the same question — on real data, they can produce two completely different business decisions. Everything in Section 18 exists to close that gap. Rules 1–3 alone — schema, business rules, edge cases — turned five wrong names into one right one.

References

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